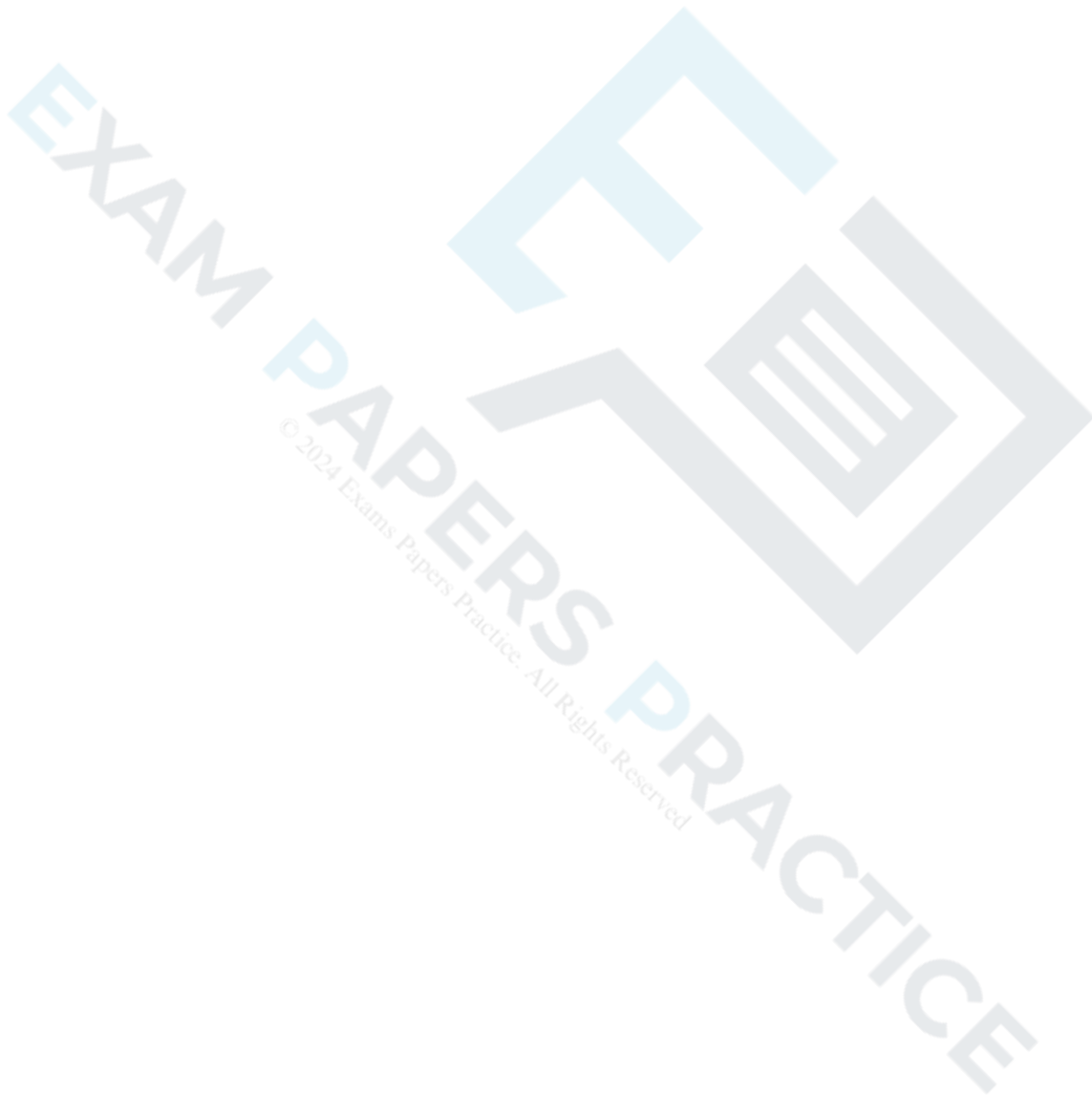


Q1.

Evaluate the improper integral $\int_0^{\infty} \frac{4x - 30}{(x^2 + 5)(3x + 2)} dx$, showing the limiting process used.

Give your answer as a single term.

(Total 8 marks)



Q1.

Marking Instructions	AO	Marks	Typical Solution
Splits integrand into partial fractions of the form $\frac{Ax+B}{x^2+5} + \frac{C}{3x+2}$	AO3.1a	M1	$\frac{4x-30}{(x^2+5)(3x+2)} \equiv \frac{Ax+B}{x^2+5} + \frac{C}{3x+2}$
Sets up an identity from which to solve for A, B and C	AO1.1a	M1	$\Rightarrow 4x - 30 \equiv (Ax + B)(3x + 2) + C(x^2 + 5)$
Obtains correct values of A, B and C CAO	AO1.1b	A1	Compare coefficients of x : $4 = 2A + 3B$
Integrates 'their' two terms correctly FT provided both M1 marks awarded	AO1.1b	A1F	Compare coefficients of x^2 : $0 = 3A + C$
Applies the laws of logs to 'their' integral correctly	AO1.1a	M1	Compare constant terms: $-30 = 2B + 5C$
Applies limits (a and 0) to 'their' integral correctly	AO1.1a	M1	$\therefore -30 = 2B - 15A$ $-30 = 2B - 15\left(\frac{4-3B}{2}\right)$
Shows the limiting process used with clear detailed working	AO2.1	R1	$B = 0$ $A = 2$ $C = -6$
Obtains correct single term solution CAO	AO1.1b	A1	$\int \frac{4x-30}{(x^2+5)(3x+2)} dx$ $= \int \frac{2x}{x^2+5} - \frac{6}{3x+2} dx$ $= \ln(x^2+5) - 2\ln(3x+2) + c$ $\int_0^a \frac{4x-30}{(x^2+5)(3x+2)} dx$ $= \lim_{a \rightarrow \infty} \int_0^a \frac{2x}{x^2+5} - \frac{6}{3x+2} dx$ $= \lim_{a \rightarrow \infty} \left[\ln(x^2+5) - 2\ln(3x+2) \right]_0^a$ $= \lim_{a \rightarrow \infty} \left[\ln\left(\frac{x^2+5}{(3x+2)^2}\right) \right]_0^a$ $= \lim_{a \rightarrow \infty} \left[\ln\left(\frac{a^2+5}{9a^2+12a+4}\right) - \ln\left(\frac{5}{4}\right) \right]$ $= \ln\left(\frac{1}{9}\right) - \ln\left(\frac{5}{4}\right)$

			$= \ln\left(\frac{4}{45}\right)$
Total 8 marks			

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